

Ambarish S. Ghatpande, PhD

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Google Scholar: <https://scholar.google.com/citations?hl=en&user=wXXa7wMAAAAJ>

Academic Profile

I am a biophysicist and neuroscientist with experience spanning ion channel biophysics, neuromodulation, sensory circuit physiology, and population-level neural dynamics. My recent work has centered on large-scale extracellular recordings from human cortical and hippocampal organoids, both in vitro and following transplantation. I have developed end-to-end pipelines for silicon probe recordings, spike sorting (Kilosort4), and computational analysis of population activity using open source Python-based tools and adhere to best practices in AI assisted scientific coding.

My early work focused on sodium and cyclic nucleotide-gated ion channels, combining electrophysiology with chemical biology approaches to elucidate mechanisms of channel gating and pharmacological modulation. I subsequently investigated cholinergic and nitric oxide signaling in olfactory bulb circuits, contributing to an understanding of how neuromodulators shape sensory processing.

Education

PhD, Molecular Biophysics, Indian Institute of Science, Bangalore (1993–2000)

M.Sc., Zoology, Savitribai Phule Pune University (1991–1993)

B.Sc., Microbiology, Savitribai Phule Pune University (1988–1991)

Academic & Research Appointments

Senior Research Investigator, Department of Neurosurgery, University of Pennsylvania (2023–2025)

Research Associate, University of Maryland, Baltimore (2011–2012)

Research Associate, Monell Chemical Senses Center, Philadelphia (2005–2011)

Postdoctoral Fellow, University of Colorado, Denver (1999–2005)

Non-Academic Appointments

Surgical Neurophysiologist II & III, Sentient Medical Systems & SpecialtyCare (2015–2021)

Peer-Reviewed Publications (Selected)

1. Ghatpande A.S., Sikdar S.K. (1997). Competition for binding between veratridine and KIFMK. *Journal of Membrane Biology*.
2. Ghatpande A.S., Sikdar S.K. (1999). Voltage-dependent gating of veratridine-modified sodium channels. *Pflügers Archiv*.
3. Ghatpande A.S. et al. (2006). Store calcium mediates cholinergic effects on mIPSCs. *Journal of Neurophysiology*.
4. Ghatpande A.S., Gelperin A. (2009). Presynaptic muscarinic receptors enhance glutamate release. *Journal of Neurophysiology*.
5. McQuade L.E. et al. (2010). Visualization of nitric oxide production in olfactory bulb. *PNAS*.
6. Ghatpande A.S., Reisert J. (2011). Olfactory receptor neuron responses and rapid odor sampling. *Journal of Physiology*.

Conference Presentations & Abstracts

Cell Symposia: Engineering Development and Disease in Organoids (2024) – Poster

NIH BRAIN Initiative Conference (2024) – Poster

Society for Neuroscience (2008) – Poster

Teaching & Mentorship

Mentored graduate students, postdoctoral fellows, and technicians in electrophysiology and data analysis.

Teaching workshop instructor, National Centre for Biological Sciences, Bangalore.

Honors & Awards

Research Fellowship, Council for Industrial and Scientific Research, India (1993–1999)

Employee Service Award, Monell Chemical Senses Center (2010)

5-Year Service Award, SpecialtyCare (2020)

Research Skills

Electrophysiology (in vitro & in vivo), Silicon probe recordings, SpikeInterface, Kilosort, Python, MATLAB, Neural population dynamics, Organoid neuroscience, Optogenetics